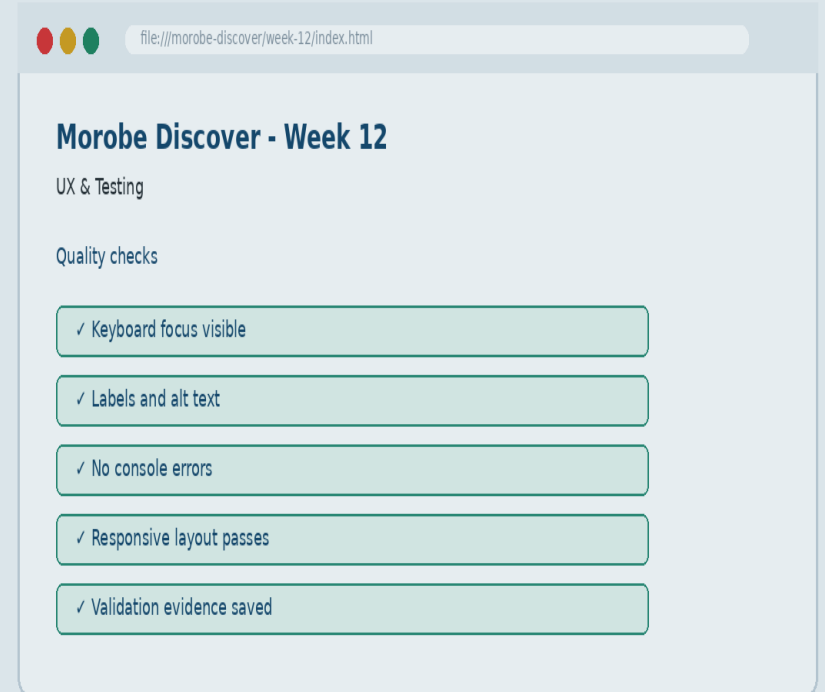


Week 12

UX & Testing

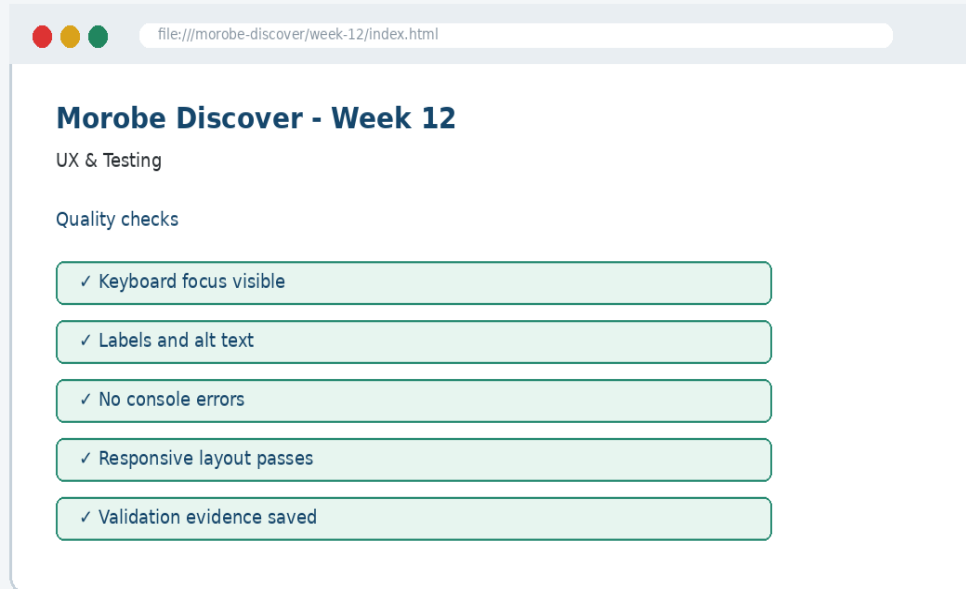
Ensure the site is accessible, validated and optimized for performance.



Learning outcomes

Ensure the site is accessible, validated and optimized for performance.
Explain the core programming/design concept for this week
Read and modify key code patterns
Test the weekly project increment
Prepare evidence for GitHub submission

Today's problem



Why it matters

Professional release requires evidence: tests, screenshots, validation reports and fixes.

Project before -> after

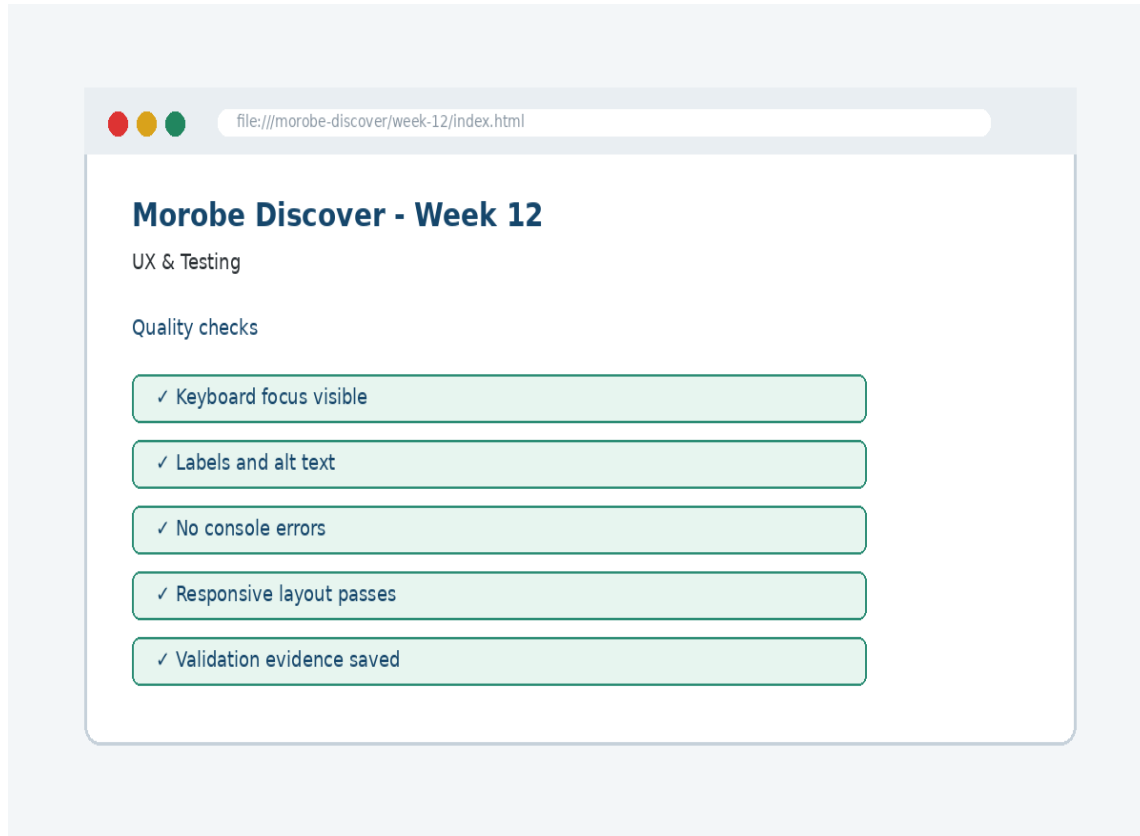
Before Week 12

Week 11 completed dynamic data rendering and filter behaviour.

After Week 12

The site has documented quality evidence: validation, accessibility, responsiveness and performance checks.

What we build this week



Create a testing checklist
Validate HTML and CSS
Test keyboard navigation and focus
Run responsive and performance checks
Record fixes in a test report

Core principle

Professional release requires evidence: tests, screenshots, validation reports and fixes.

Keep in mind

A good web developer can point to the code, the browser result, and the user reason behind the decision.

Key concepts

Concept 1

Testing checks what users actually do

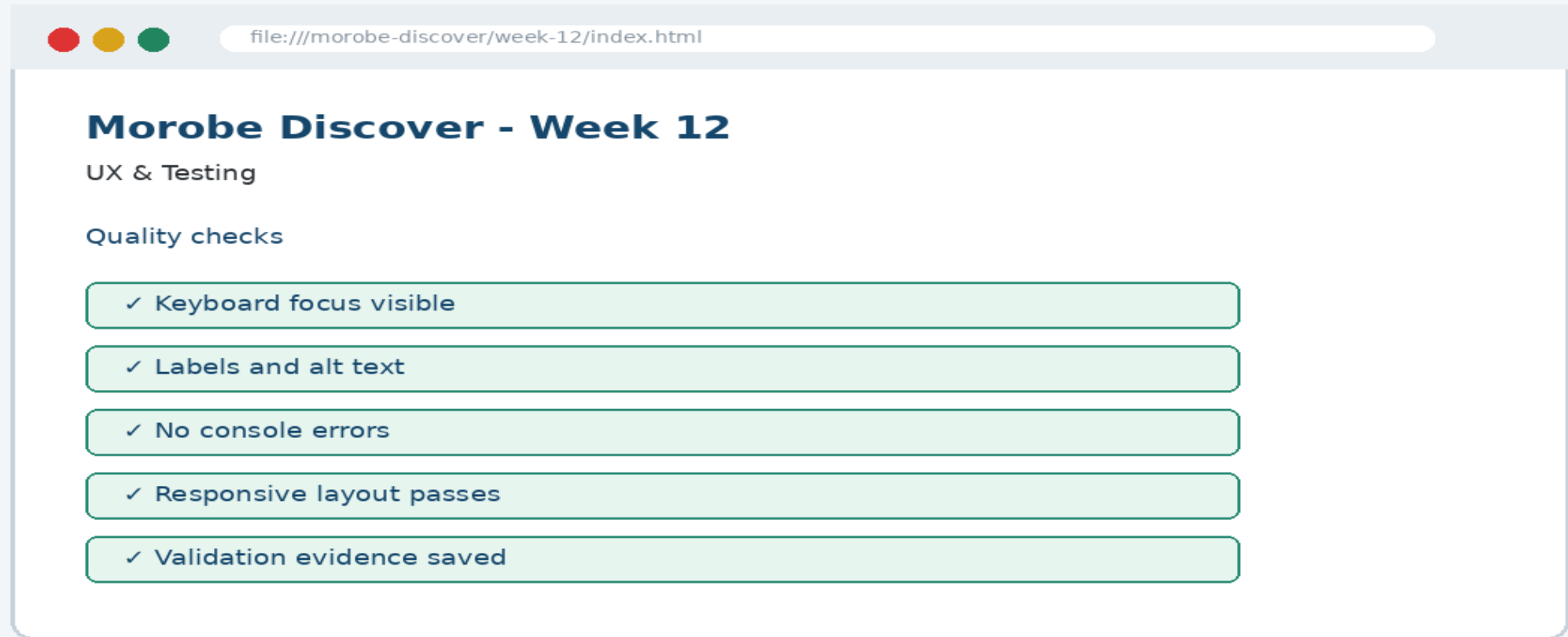
Concept 2

Validation finds code errors that browsers may hide

Concept 3

Performance and accessibility are quality requirements

Project anatomy



Key code pattern A

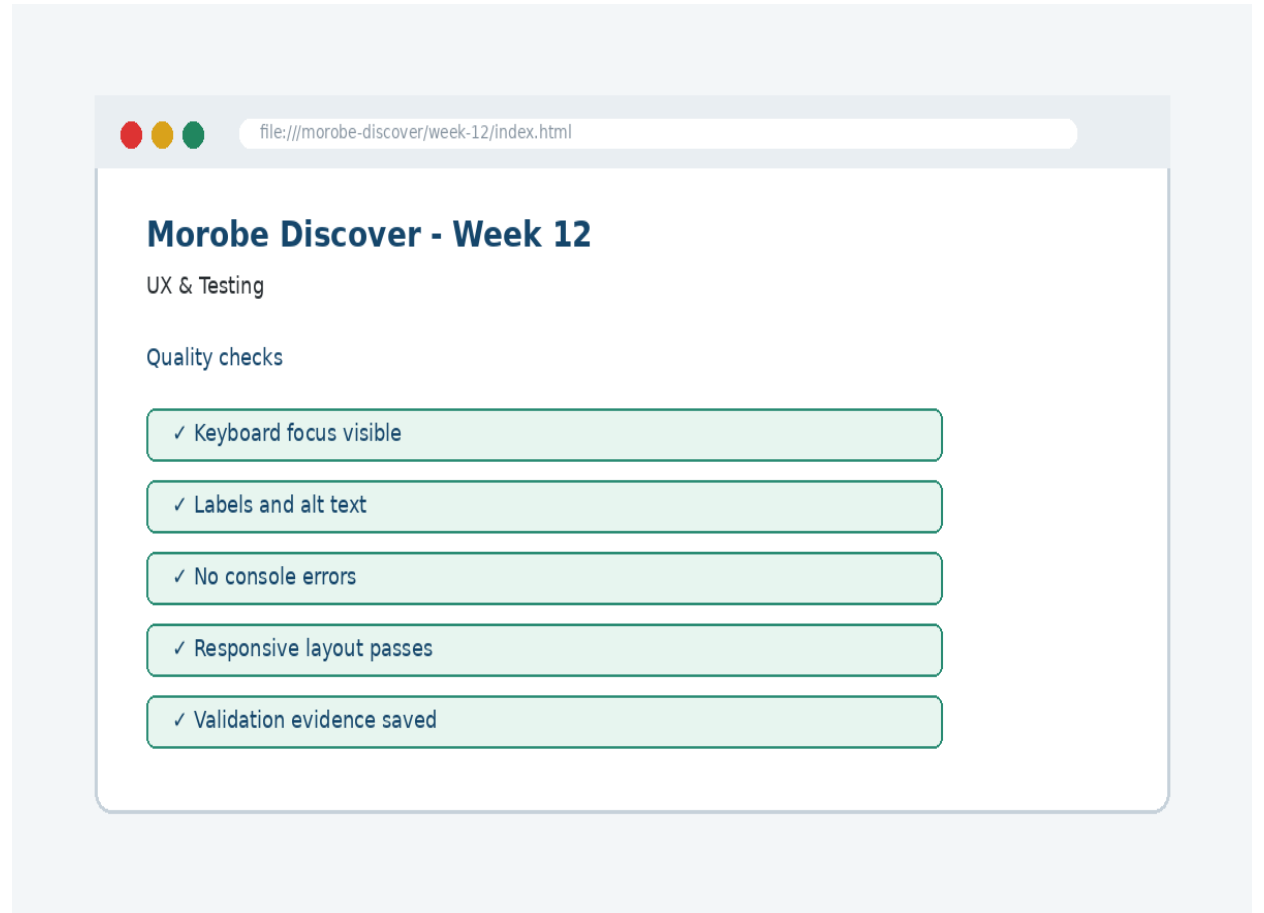
```
// Test note example  
Expected: menu opens with keyboard.  
Actual: focus moves to first link.  
Result: Pass.
```

Explain it

Which file contains this code?
Which line creates the visible change?
What should appear in the browser?

Code -> visual result

```
// Test note example  
Expected: menu opens with keyboard.  
Actual: focus moves to first link.  
Result: Pass.
```



What changed?

Project file changed for UX & Testing

The browser output is now closer to the required weekly evidence

The change supports a user task or quality requirement

The change can be tested and committed

Key code pattern B

```

```

Why this pattern is needed

The second pattern usually applies the same idea to a different component, state, layout or data flow.

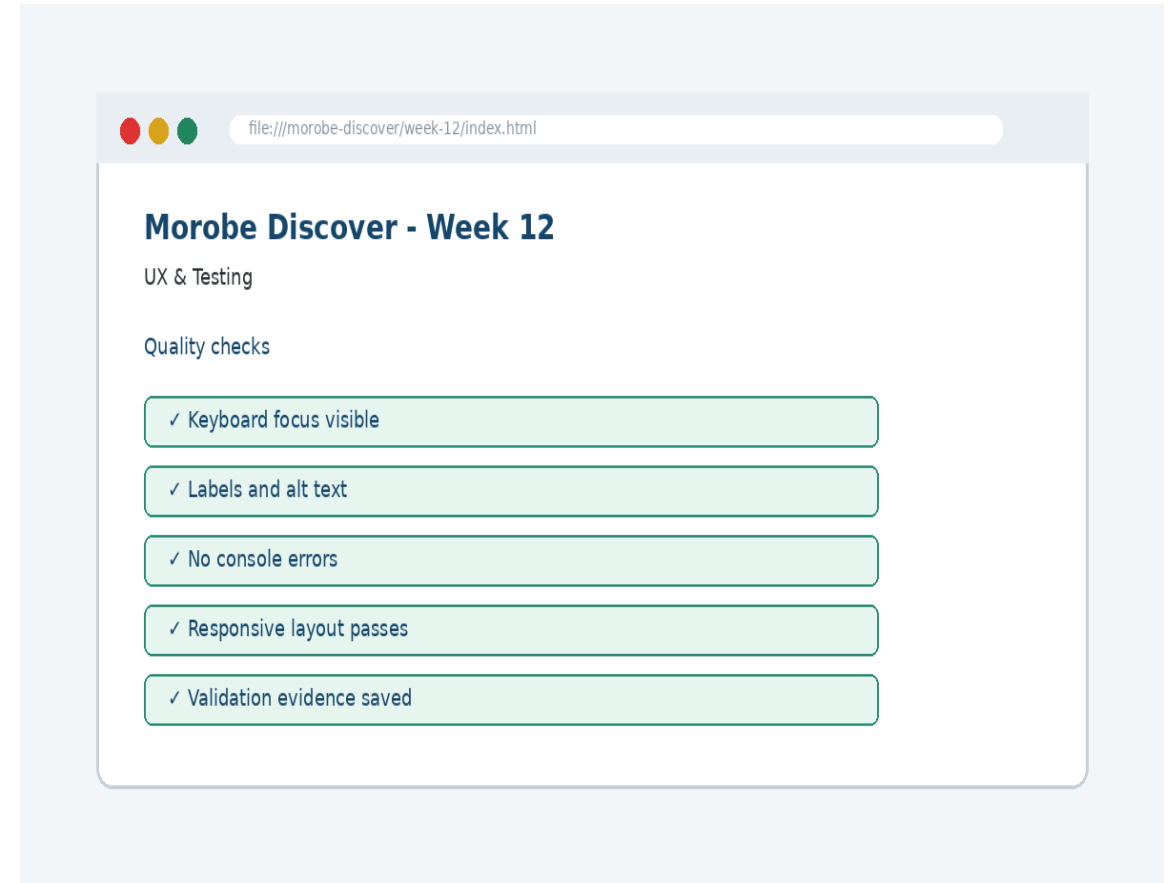
Apply it to Morobe Discover

Create a testing checklist

Validate HTML and CSS

Test keyboard navigation and focus

Run responsive and performance checks



Compare the approaches

Weak approach

Memorise the definition
Copy code without understanding
No browser evidence
No test note

Professional approach

Explain the concept
Apply the correct code pattern
Inspect the result
Commit and describe the evidence

Common mistake

Mistake

The page appears unchanged, but the student assumes the feature is complete.

Correction

Check the correct file, refresh the browser, inspect DevTools, and verify the expected output.

Why does this fail?

```
// Broken checkpoint
// File changed, but the browser result does not
change.
// Possible causes:
// - wrong file
// - file not linked
// - syntax error
// - browser not refreshed
```

Class question

Which cause would you check first, and why?

Fix the problem

- Confirm the browser is opening the correct file
- Check links to CSS/JS/data files
- Inspect the console for errors
- Use a small working example
- Test again before committing

Class activity

Activity - 7 minutes

Inspect the project and list the three most important issues to fix before deployment.

Work in pairs

Point to the project evidence

Prepare one answer to share

Implementation steps 1-3

Create a testing checklist

Validate HTML and CSS

Test keyboard navigation and focus

Implementation steps 4-5

Run responsive and performance checks
Record fixes in a test report

LIVE DEMO

Demo checklist

1. Open the current project
2. Locate the affected file
3. Add or modify the key code
4. Refresh the browser
5. Inspect the result
6. Explain the change



file:///morobe-discover/week-12/index.html

Morobe Discover - Week 12

UX & Testing

Quality checks

✓ Keyboard focus visible

✓ Labels and alt text

✓ No console errors

✓ Responsive layout passes

✓ Validation evidence saved

Predict the output

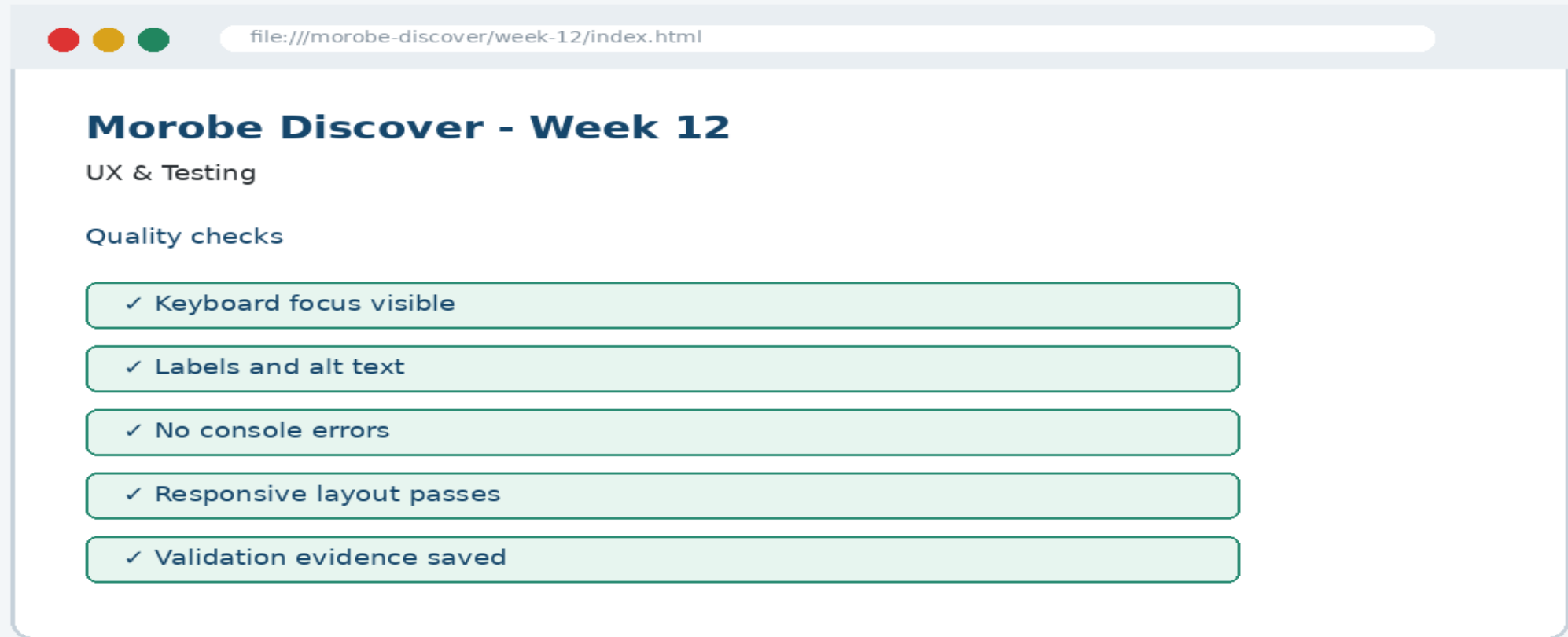
```

```

Prediction prompt

What should change on screen when this code is applied correctly?

Browser result



Testing the feature

- Verify broken links
- Check console errors
- Run Lighthouse or manual accessibility checks
- Retest after fixing problems

Inspect with browser tools

Inspection path

Right click -> Inspect
Elements panel -> structure/classes
Console -> JavaScript errors
Network/Application -> data/storage when relevant

Evidence to save

Screenshot
Short test note
Changed files
Git commit message

Git evidence

```
git add .  
git commit -m "Week 12 UX & Testing project  
increment"
```

Why it matters

Git history shows progression from Week 01 to final deployment. It is part of the assessment evidence.

Complete source code

Repository:
`https://github.com/YOUR-USERNAME/is229-morobe-discover`

Weekly folder:
`/week-12/`

Complete project:
`/complete-project/`

Reminder

Do not paste a whole file into a slide. Use the repository for complete code.

Mini challenge

Challenge

Modify one part of the Week 12 implementation and explain what changed in the browser.

State the file edited

State the line/section changed

Show the before and after result

What counts as evidence?

Technical evidence

Changed code files
Browser screenshot
Test note
Git commit

Understanding evidence

Code explanation
Why this method fits
Connection to previous week
Connection to next week

Project lab handoff

Create a testing checklist

Validate HTML and CSS

Test keyboard navigation and focus

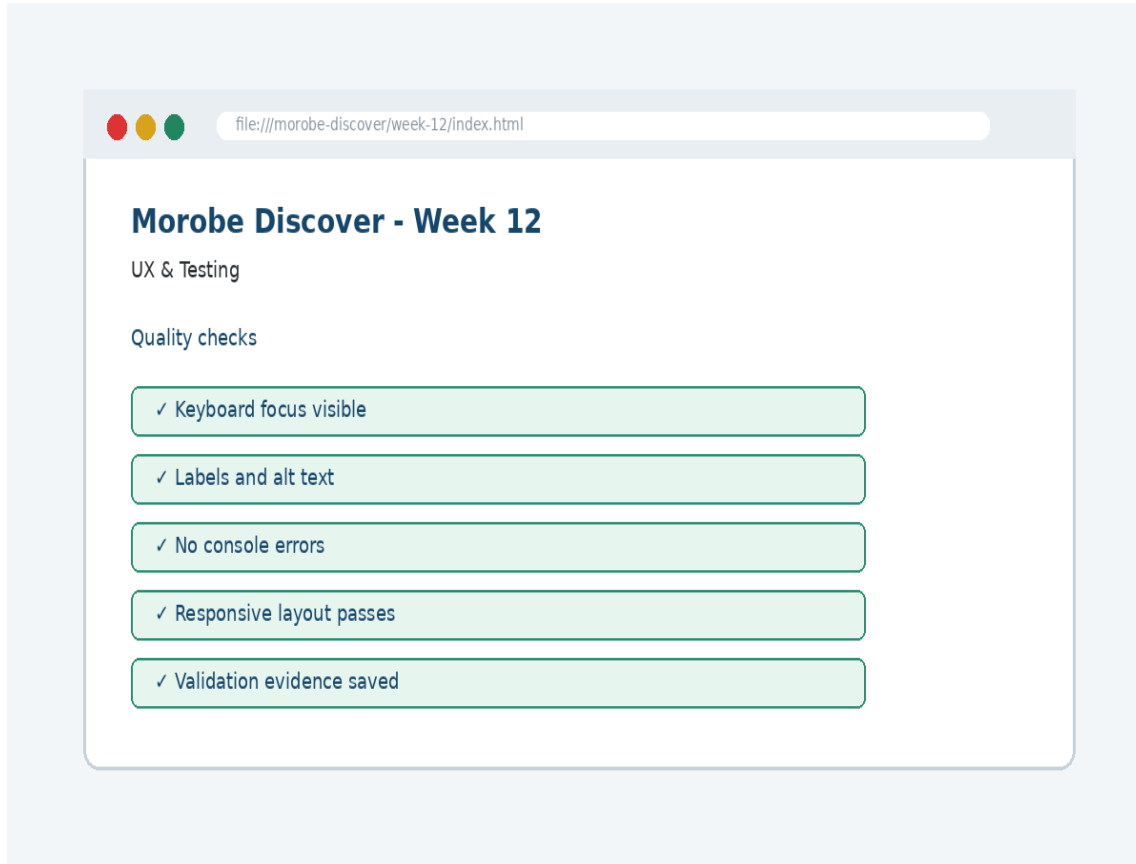
Run responsive and performance checks

Record fixes in a test report

Definition of done

Week 12 feature is implemented
Project still preserves earlier week features
Browser result matches the expected output
Test note and screenshot are saved
Git commit message is meaningful

What we built today



The site has documented quality evidence: validation, accessibility, responsiveness and performance checks. Evidence: Accessibility checks, validation reports and quality tests
Technology: Testing Tools

Lecture summary

Professional release requires evidence: tests, screenshots, validation reports and fixes.

Testing checks what users actually do

Validation finds code errors that browsers may hide

Testing and Git are part of development, not extras

Prepare for next class

Finish this week's lab before the next class
Read the next topic in advance
Bring the current project ready to extend
Be ready to explain one code decision

Exit question

How did Week 12 improve Morobe Discover, and what will the next week build on?